

Spendly

Offline-First · Zero-Knowledge · Peer-to-Peer Finance Ecosystem

Student Name · Enrollment Number · Semester VII · Information
Technology · College Name



The Problem

Finance today demands trust in systems designed to fail you.

Privacy Risks

Your data is the product

Cloud Dependency

Single points of failure

Internet Required

No connectivity, no finance

Centralized Databases

Breaches are inevitable

Monthly POS Costs

Hardware is a barrier

Introducing Spendly

Three applications. One ecosystem. Zero compromise.

Spendly

Personal finance & expense tracking — offline-first, zero-knowledge

Spendly Shop

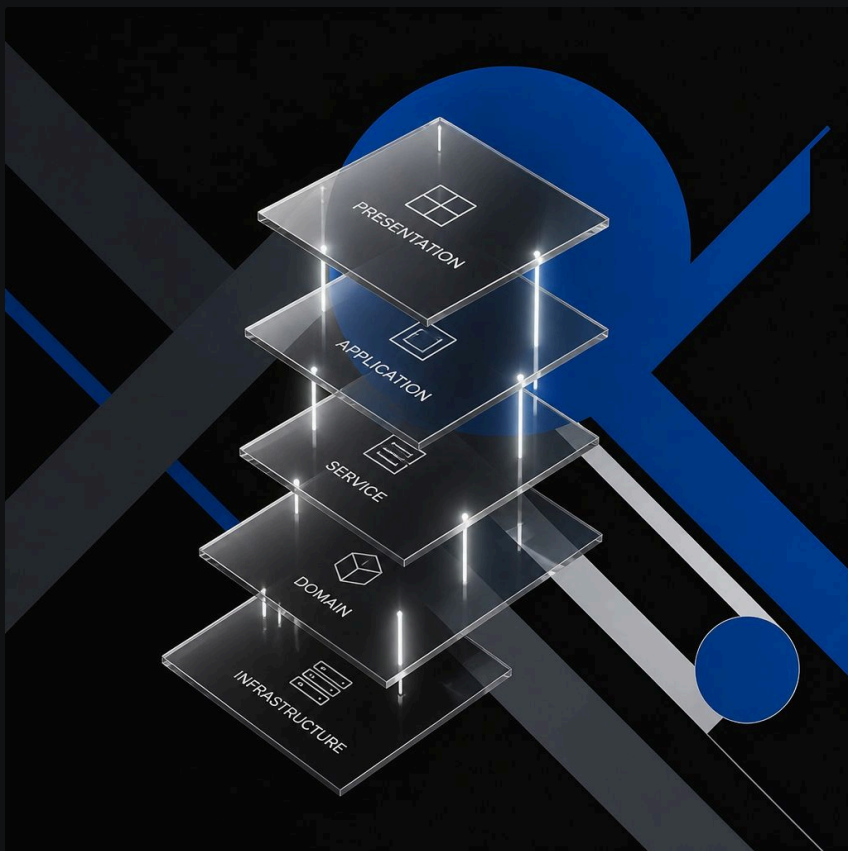
Merchant billing, inventory & QR invoices — no POS hardware

Spendly Radar

Local offer discovery & peer-to-peer coupon sharing



System Architecture



01

React + PWA

Progressive Web App shell

02

IndexedDB + Dexie.js

Offline-first local database

03

WebRTC + PeerJS

Direct peer-to-peer networking

04

Cloudflare Workers + KV

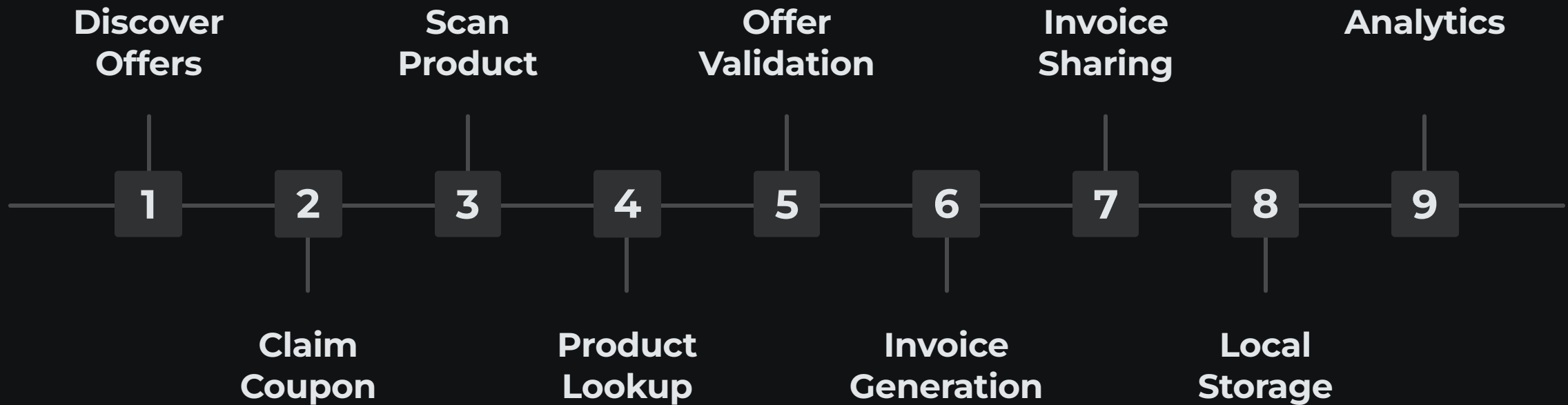
Edge compute & key-value store

05

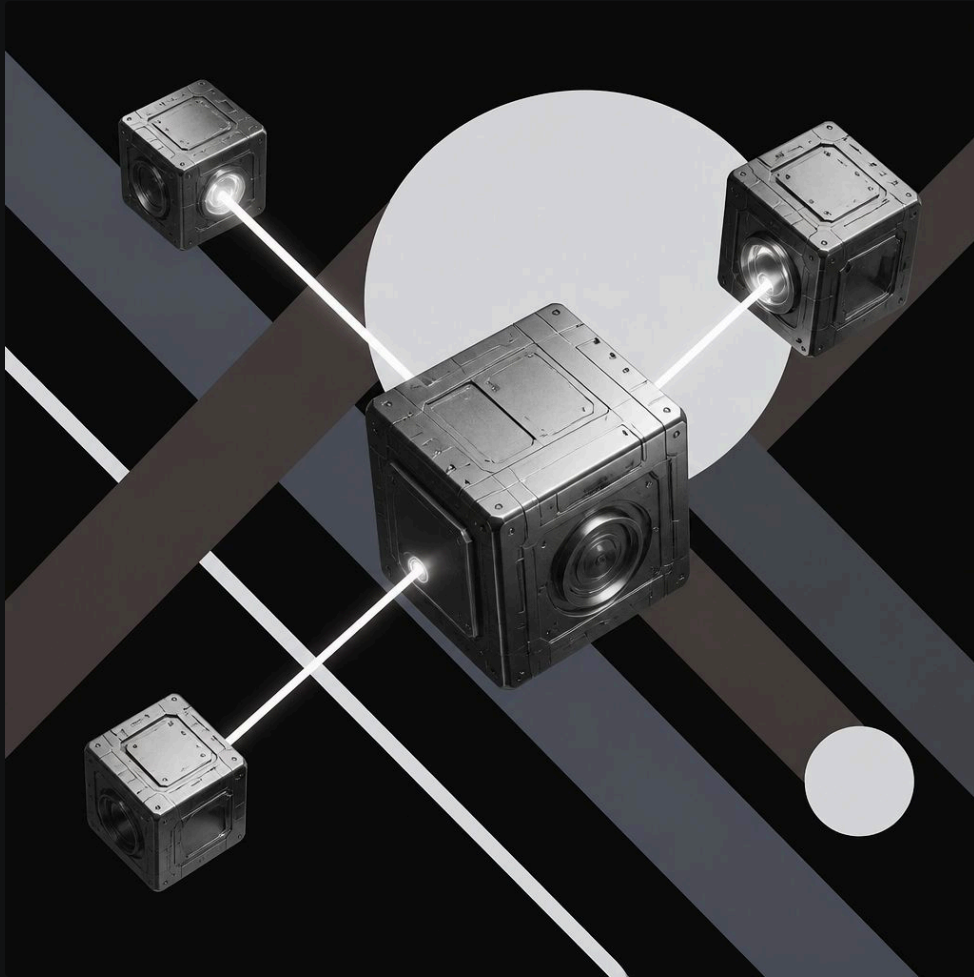
WebAuthn + AES-256-GCM

Biometric auth & end-to-end encryption

Working Flow



Technology Stack



Frontend

React · Vite · Tailwind CSS ·
Zustand · Framer Motion

Backend

Cloudflare Workers · Worker
KV

Networking

WebRTC · PeerJS · ntfy.sh

Security

WebAuthn · AES-256-GCM

Database

IndexedDB · Dexie.js

Security by Design

Your data never leaves your device. No exceptions.



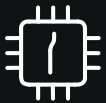
Zero-Knowledge

Server knows nothing



WebAuthn

Biometric unlock



Secure Enclave

Hardware-backed isolation



Offline-First

Works without internet



AES-256-GCM

Military-grade encryption



IndexedDB

Encrypted local storage

Key Features



Expense Tracking



Merchant Billing



Inventory



Cash Wallet



EMI Reminder



QR Invoice



Offline Sync



Analytics

Advantages & Future Scope



TODAY

Advantages

- Offline-first architecture
- Zero server cost
- Privacy by design
- Fast billing
- Secure communication

TOMORROW

Future Scope

- AI analytics
- Voice billing
- NFC payments
- UPI integration
- GST & multi-store

Spendly

A privacy-first, offline financial ecosystem built on zero-knowledge architecture and peer-to-peer communication.

Thank You Questions & Answers

